MATH 111 Calculus 1

The derivative as a function

Reading f' off f · where the derivative fails to exist

Week 3 · Lecture 8 · Stewart §2.2

Dr. Salnikova Chekeres · New Uzbekistan University · Fall 2026

While you wait

Sit with your team. In the meantime: sketch $y = x^2$. Now, under it, sketch the *slope* of that curve at each point, as a new graph. What shape do you get?

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"Does every function have a derivative everywhere?"

No. The derivative is a limit, and limits can fail to exist. We will find exactly three ways a graph can refuse to have a tangent — and one of them will surprise you.

STAND UP

"If f is continuous at a, then f has a derivative at a."

Stand up if you think this is TRUE.

Stay standing until I say "sit". Then find someone with the other opinion — 60 seconds.

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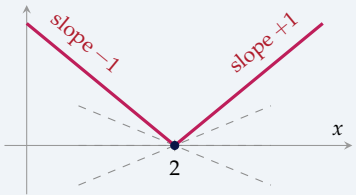
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Verdict

FALSE. $f(x) = |x - 2|$ is continuous at 2 — you can draw it without lifting the pen — yet it has no tangent there: coming from the left the slope is -1, from the right it is $+1$, and the difference quotient has no limit. Continuity is about the values; differentiability is about the direction.



$y = |x - 2|$: which dashed line is *the* tangent?

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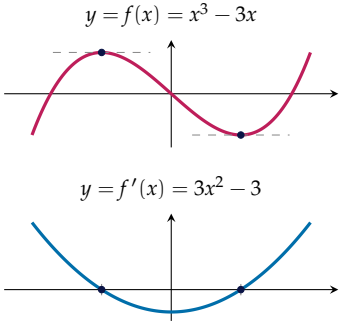
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$$= 3x^2 - 3 \quad \text{— for } \textit{every } x\text{, in one computation.}$$

Reading f' off the picture of f

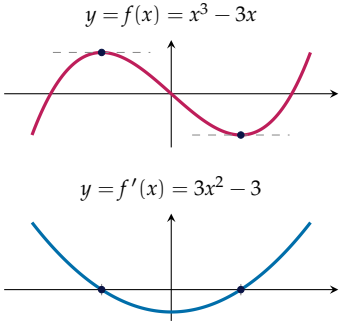
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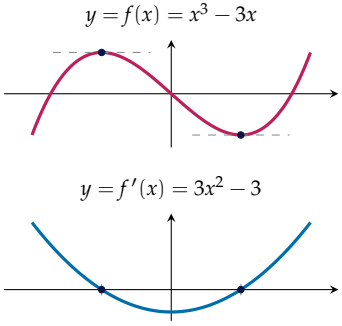
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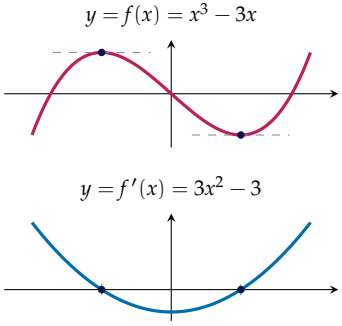
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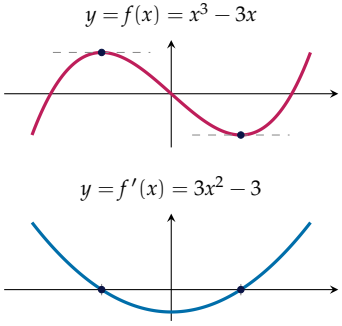
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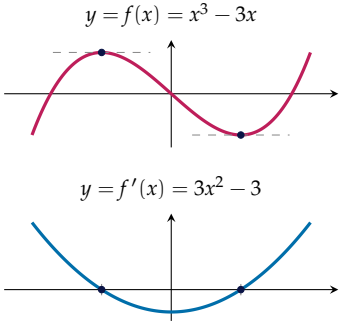


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Below: $f(x) = x^3 - 3x$ and, beneath it, $f'(x) = 3x^2 - 3$. The two zeros of f' sit exactly under the two flat spots of f.



Notation — four ways to write the same thing

The same object

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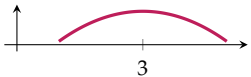
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Why two notations survive

f' is short and says "a new function"; $\dfrac{dy}{dx}$ carries the units and the variables with it, which is why the chain rule will prefer it.

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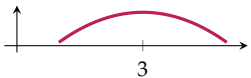
D. $f'(x) > 0$ for every x.

In your team

Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

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60 sec

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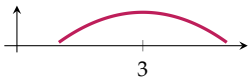
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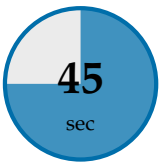


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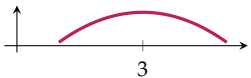


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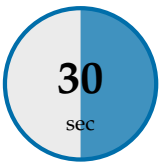


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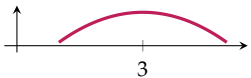


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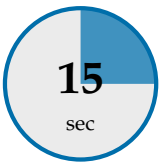


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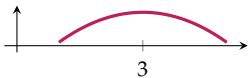


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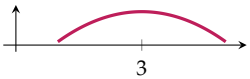
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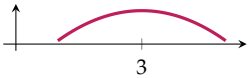
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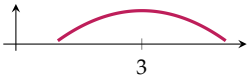
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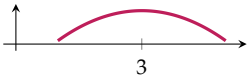
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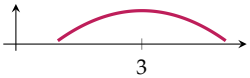
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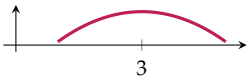
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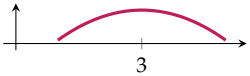
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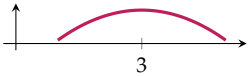
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Answer

B. At the top the tangent is horizontal, so $f'(3) = 0$; left of it the curve rises ($f' > 0$), right of it it falls ($f' < 0$). A confuses *the biggest value of f* with *the biggest value of f'*: at the top f is largest and f' is zero. C would need a corner. D forgets the falling half.

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Let $x \to a$. The first factor tends to $f'(a)$ (it exists, by hypothesis) and the second to 0, so by the product law $\lim_{x \to a} \big[f(x) - f(a)\big] = f'(a) \cdot 0 = 0$.

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Proof. For $x \neq a$ we may write $f(x) - f(a) = \dfrac{f(x) - f(a)}{x - a} \cdot (x - a)$.

Let $x \to a$. The first factor tends to $f'(a)$ (it exists, by hypothesis) and the second to 0, so by the product law $\lim_{x \to a} \big[f(x) - f(a)\big] = f'(a) \cdot 0 = 0$.

Hence $\lim_{x \to a} f(x) = f(a)$: continuity. □ Smooth ⇒ unbroken.

Differentiable — and what it costs

Definition

f is differentiable at a if $f'(a)$ exists; differentiable on an interval if it is differentiable at every number in it.

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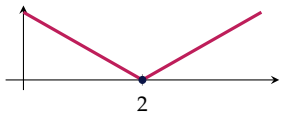
Hence $\lim_{x \to a} f(x) = f(a)$: continuity. □ Smooth $\Rightarrow$ unbroken.

The converse is false — and the warm-up already showed why.

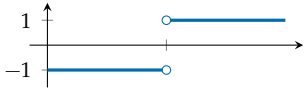
$|x-2|$ at its corner, in full

$f(x) = |x-2|$. At $a = 2$ the difference quotient is
$\dfrac{f(2+h)-f(2)}{h} = \dfrac{|h|-0}{h} = \dfrac{|h|}{h}$.

$y = |x-2|$: a *corner*



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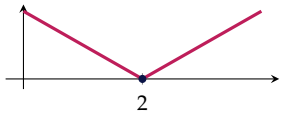


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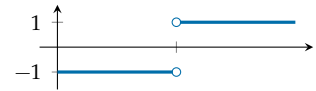
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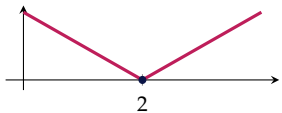
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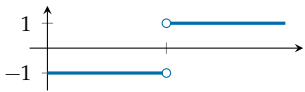
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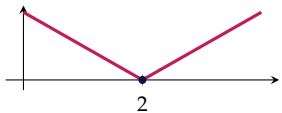
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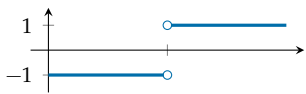
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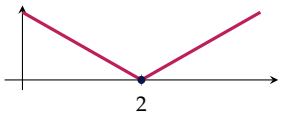
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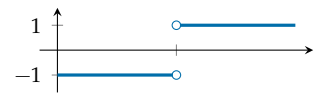
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Away from 2: $f'(x) = 1$ for $x > 2$, $f'(x) = -1$ for $x < 2$. Domain of f: $\mathbb{R}$; of f': $\mathbb{R} \setminus \{2\}$.

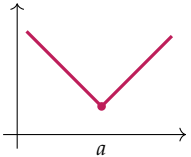
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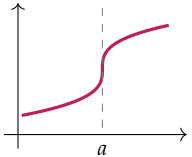
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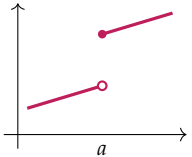
Three ways to have no derivative



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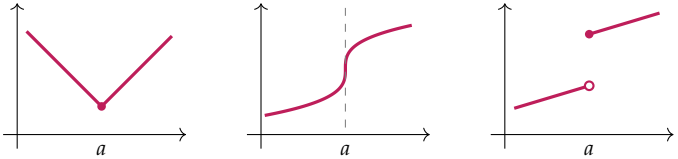


vertical tangent



discontinuity

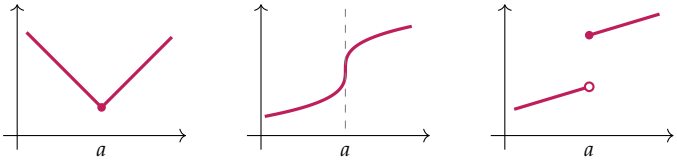
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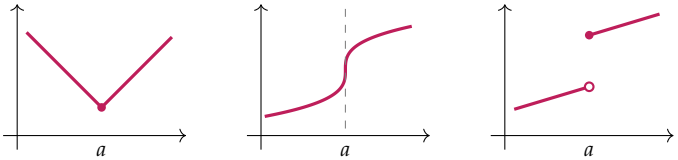


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Vertical tangent: the difference quotient runs to $\pm\infty$. There *is* a tangent line, $x = a$; it has no slope. Example: $y = \sqrt[3]{x}$ at 0.

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Discontinuity: by the theorem, no continuity means no derivative. A jump, a hole or a blow-up all kill it.

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What the second derivative measures

If $s(t)$ is position, then $s'(t) = v(t)$ is velocity and $s''(t) = a(t)$ is acceleration — the rate at which the rate is changing. (The third derivative of position has a name too: *jerk*, the rate of change of acceleration, in $\mathrm{m/s^3}$ — the lurch you feel when a bus starts or brakes suddenly.)

Question 18

Let $f(x) = |x + 4|$. Which statement is true?

A. f is not continuous at -4.

B. f is continuous at -4 and $f'(-4) = 0$.

C. f is continuous at -4, but $f'(-4)$ does not exist.

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Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

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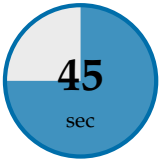
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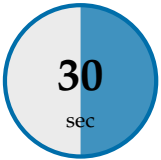
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Eyes on the screen. Hands up when your letter appears — hold for three seconds.

A B C D

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Answer

C. $\lim_{x \to -4} |x + 4| = 0 = f(-4)$, so f is continuous at -4. But the difference quotient is $\frac{|h|}{h}$ again: $+1$ from the right, -1 from the left, so $f'(-4)$ does not exist — the graph has a corner at -4. B is the trap: at the lowest point of the graph students expect a horizontal tangent, but there is no tangent at all. D takes only the right-hand side.

STAND UP

"There is a function that is continuous at every point of $\mathbb{R}$ and differentiable at no point of $\mathbb{R}$."

Stand up if you think this is TRUE.

Stay standing until I say "sit". Then find someone with the other opinion — 60 seconds.

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Verdict

TRUE. A corner at one point is easy; a corner at *every* point sounds impossible, and for two centuries most mathematicians assumed it was. Weierstrass exhibited one in 1872 — a sum of infinitely many ever smaller, ever faster cosine waves. It is unbroken everywhere and has a direction nowhere. Your intuition that "continuous means you could draw it" survives; the intuition that "drawable means smooth in most places" does not.

A little history · 1806–1916

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$$W(x) = \sum_{n \geq 0} a^n \cos(b^n \pi x), \qquad 0 < a < 1, \;\; b \text{ an odd integer}, \;\; ab > 1 + \tfrac{3\pi}{2} :$$

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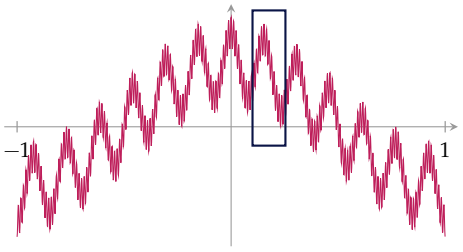
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Hermite wrote to Stieltjes in 1893: *"I turn away with fear and horror from the lamentable plague of continuous functions which do not have derivatives."*
Poincaré called them *monsters*. Once pictures could lie, only the inequalities could be trusted.

The monster, drawn

Take $a = \frac{1}{2}$ and $b = 13$: $W(x) = \sum_{n=0}^{\infty} \frac{\cos(13^n \pi x)}{2^n} = \cos \pi x + \frac{1}{2} \cos 13\pi x + \frac{1}{4} \cos 169\pi x + \cdots,$

and $ab = 6.5 > 1 + \frac{3\pi}{2} \approx 5.71$.

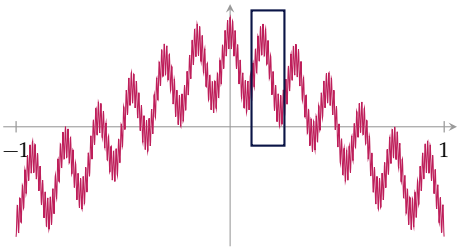


three terms on $[-1, 1]$; the box is $2/13$ wide

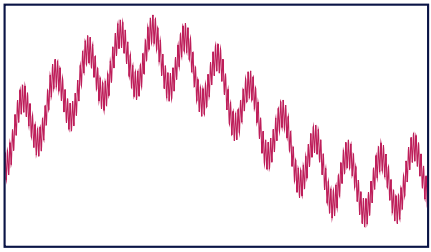
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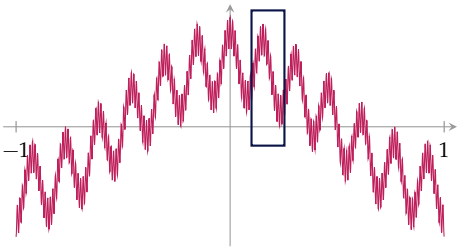


inside the box, 13 times enlarged, with four terms

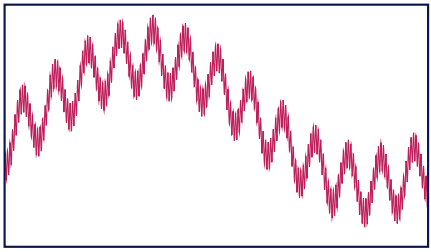
The monster, drawn

Take $a = \frac{1}{2}$ and $b = 13$: $W(x) = \sum_{n=0}^{\infty} \frac{\cos(13^n \pi x)}{2^n} = \cos \pi x + \frac{1}{2} \cos 13\pi x + \frac{1}{4} \cos 169\pi x + \cdots,$

and $ab = 6.5 > 1 + \frac{3\pi}{2} \approx 5.71$.



three terms on $[-1, 1]$; the box is $2/13$ wide



inside the box, 13 times enlarged, with four terms

The same kind of picture at every scale: each term adds 13 times more wiggles, half as tall. A tangent needs the wiggles to flatten out as you zoom in — here they never do.

The monster is continuous everywhere — an ε–δ proof

Two facts: (i) $|\cos u - \cos v| \le |u - v|$, from $\cos u - \cos v = -2 \sin \frac{u+v}{2} \sin \frac{u-v}{2}$, $|\sin| \le 1$, $|\sin t| \le |t|$; (ii) for $0 < a < 1$, $\sum_{n>N} a^n = \frac{a^{N+1}}{1-a}$ — so the series for W converges.

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Fix x_0 and $\varepsilon > 0$. Split $W = S_N + R_N$: the head $S_N(x) = \sum_{n=0}^{N} a^n \cos(b^n \pi x)$ and the tail $R_N(x) = \sum_{n>N} a^n \cos(b^n \pi x)$.

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The head moves slowly: by (i), $|S_N(x) - S_N(x_0)| \le \sum_{n=0}^{N} a^n \, b^n \pi \, |x - x_0| = C_N \, |x - x_0|$, with $C_N = \pi \sum_{n=0}^{N} (ab)^n$. Take $\delta = \dfrac{\varepsilon}{3C_N}$.

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Then $|x - x_0| < \delta$ gives

$$|W(x) - W(x_0)| \;\le\; |S_N(x) - S_N(x_0)| + |R_N(x)| + |R_N(x_0)| \;<\; \tfrac{\varepsilon}{3} + \tfrac{\varepsilon}{3} + \tfrac{\varepsilon}{3} = \varepsilon. \qquad \blacksquare$$

. . . and differentiable nowhere — Weierstrass's proof (1)

Now $0 < a < 1$, b an odd integer, $ab > 1 + \frac{3\pi}{2}$. Fix x_0. For each m write $b^m x_0 = \alpha_m + \varepsilon_m$ with α_m an integer and $-\frac{1}{2} \le \varepsilon_m < \frac{1}{2}$, and take the two points

$$y_m = \frac{\alpha_m - 1}{b^m} \;<\; x_0 \;<\; z_m = \frac{\alpha_m + 1}{b^m}, \qquad x_0 - y_m = \frac{1 + \varepsilon_m}{b^m}, \quad z_m - x_0 = \frac{1 - \varepsilon_m}{b^m}.$$

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The first m terms are tame. By $|\cos u - \cos v| \le |u - v|$:

$$|S_1| \le \textstyle\sum_{n=0}^{m-1} a^n b^n \pi = \pi \dfrac{(ab)^m - 1}{ab - 1} < \dfrac{\pi}{ab - 1} (ab)^m.$$

The other terms are wild, and all push the same way. For $n \geq m$, $b^n \pi y_m = b^{n-m}(\alpha_m - 1)\pi$ is an integer multiple of π with the parity of $\alpha_m - 1$, because b is odd; so $\cos(b^n \pi y_m) = -(-1)^{\alpha_m}$. And $\cos(b^n \pi x_0) = \cos\big(b^{n-m}\pi(\alpha_m + \varepsilon_m)\big) = (-1)^{\alpha_m}\cos(b^{n-m}\pi\varepsilon_m)$. Hence

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Together: the quotient at y_m has the sign of $(-1)^{\alpha_m}$ and absolute value $> (ab)^m\big(\frac{2}{3} - \frac{\pi}{ab-1}\big)$; at z_m the same computation gives the opposite sign and the same bound. The bracket is positive exactly because $ab > 1 + \frac{3\pi}{2}$, and $(ab)^m \to \infty$.

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So difference quotients at points closer and closer to x_0 are unbounded: they have no limit, and $W'(x_0)$ does not exist. Since x_0 was any point — nowhere. ■

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This week:

Homework: Stewart §2.2 — read and work out the examples. Next: the rules, so that we never compute a limit again.

EXIT TICKET

Tear a quarter page from a notebook — one per team.

1. Team name; names of those present today.
2. One sentence: *"f can be continuous at a without being differentiable at a because …"*
3. Your muddiest point — the one thing still unclear.
4. One moment today that surprised you.

Give it to the professor or the course assistant as you go. Thank you.